

COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No. 10

Combinational Circuit

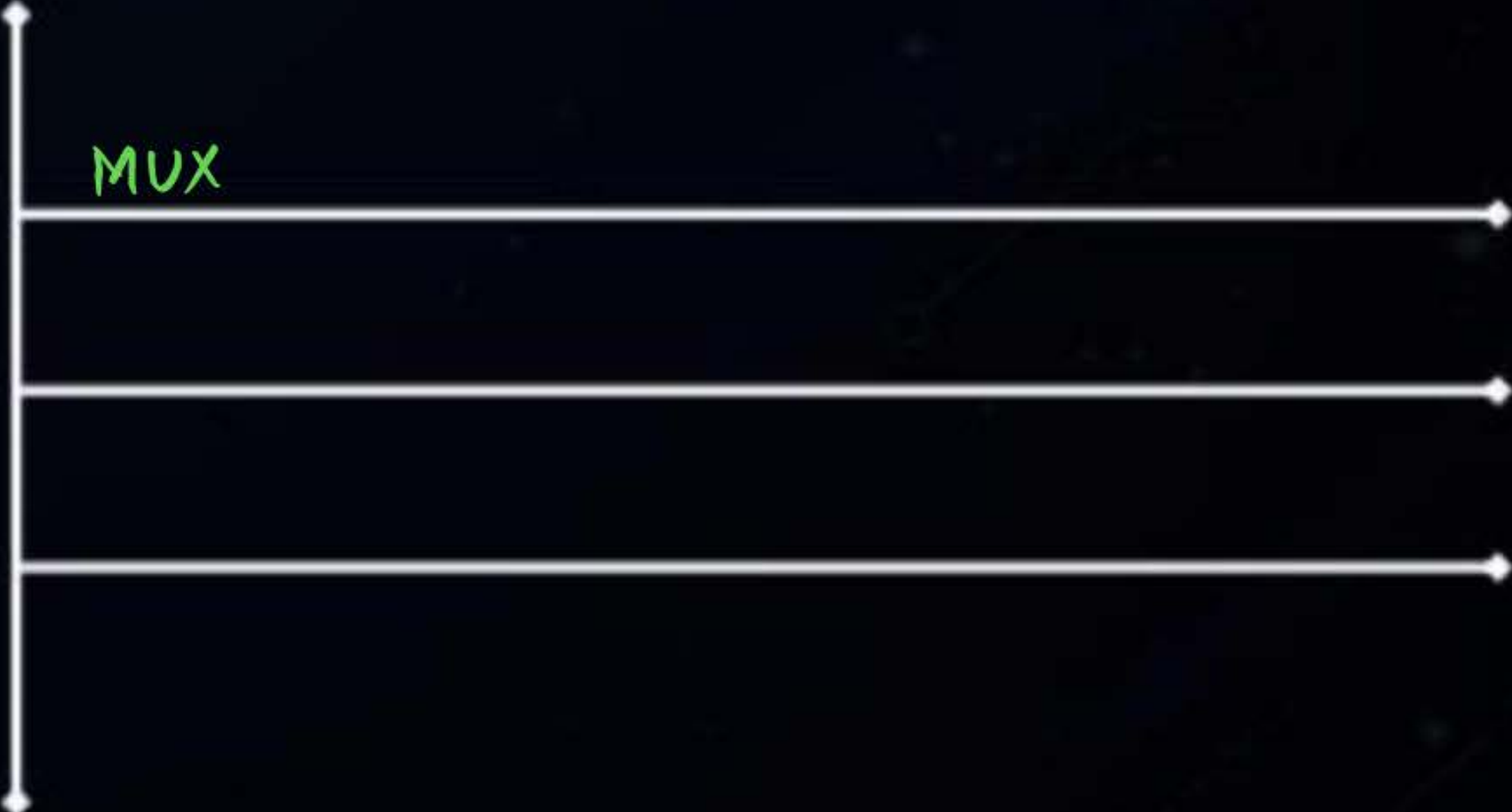


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Recap of Previous Lecture

MUX





Topics to be Covered

MUX Cont.

DeMUX

[Question 11]



A 4 : 1 multiplexer is used for generating output carry of a full adder. A and B are bits to be added while C_{in} is input carry and C_{out} is output carry. Which of the following is choice of signals at I_0, I_1, I_2, I_3 .

$$C_{out}[1, 0, C_{in}] = C_{in}, I_2 = C_{in}$$

(a) ✓ $I_0 = 0, I_1 = C_{in}, I_2 = C_{in}, I_3 = 1$

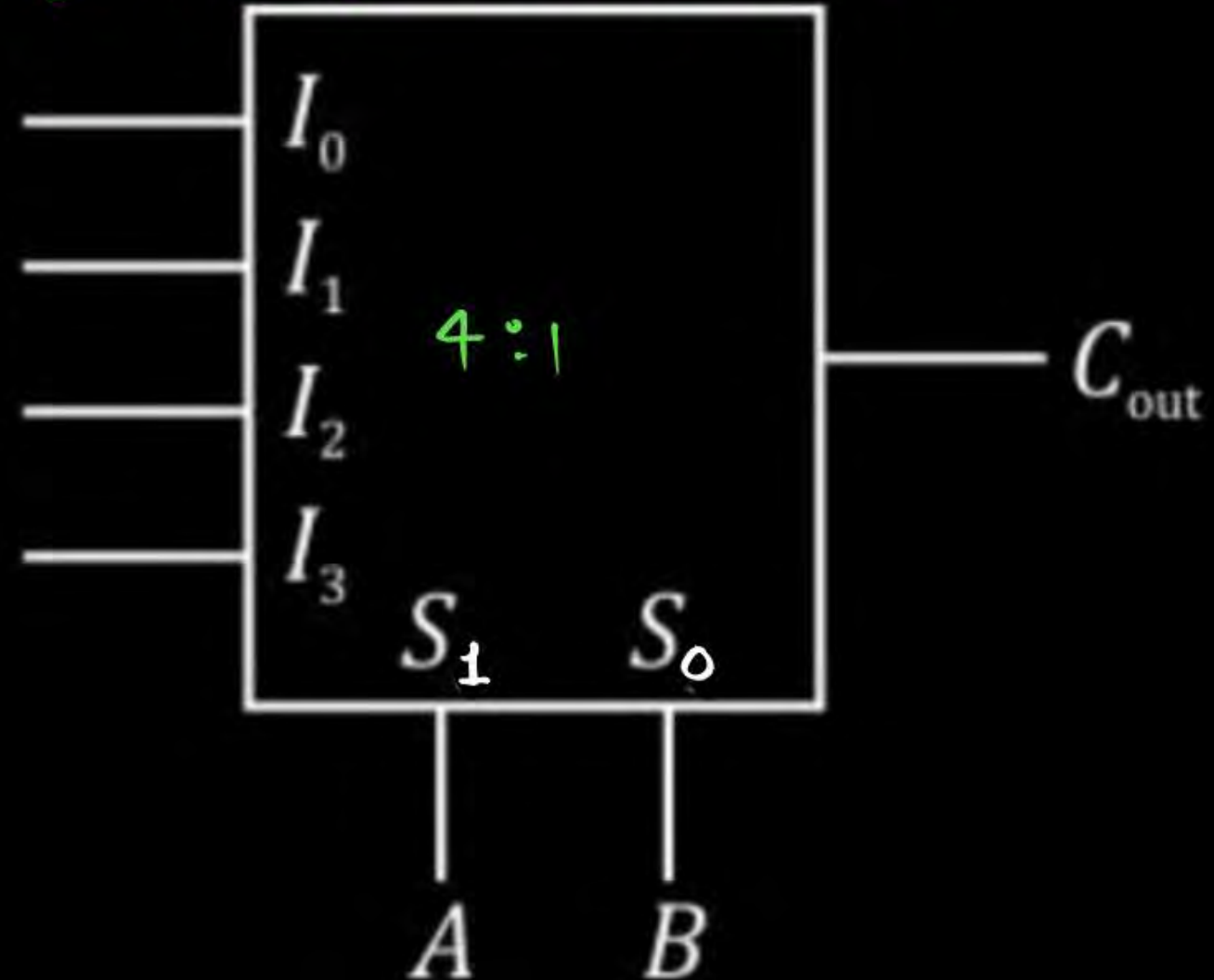
(b) ✗ $I_0 = 1, I_1 = C_{in}, I_2 = C_{in}, I_3 = 1$

(c) ✗ $I_0 = C_{in}, I_1 = 0, I_2 = 1, I_3 = C_{in}$

(d) ✗ $I_0 = 0, I_1 = C_{in}, I_2 = 1, I_3 = C_{in}$

$$C_{out}[A, B, C_{in}] = AB + BC_{in} + AC_{in}$$

$$C_{out}[0, 0, C_{in}] = 0, I_0 = 0$$



Question 12

A digital circuit is as given below :

$y(A, B, C, D)$ will be

(a) ~~$A\bar{B}C + A\bar{C}D$~~

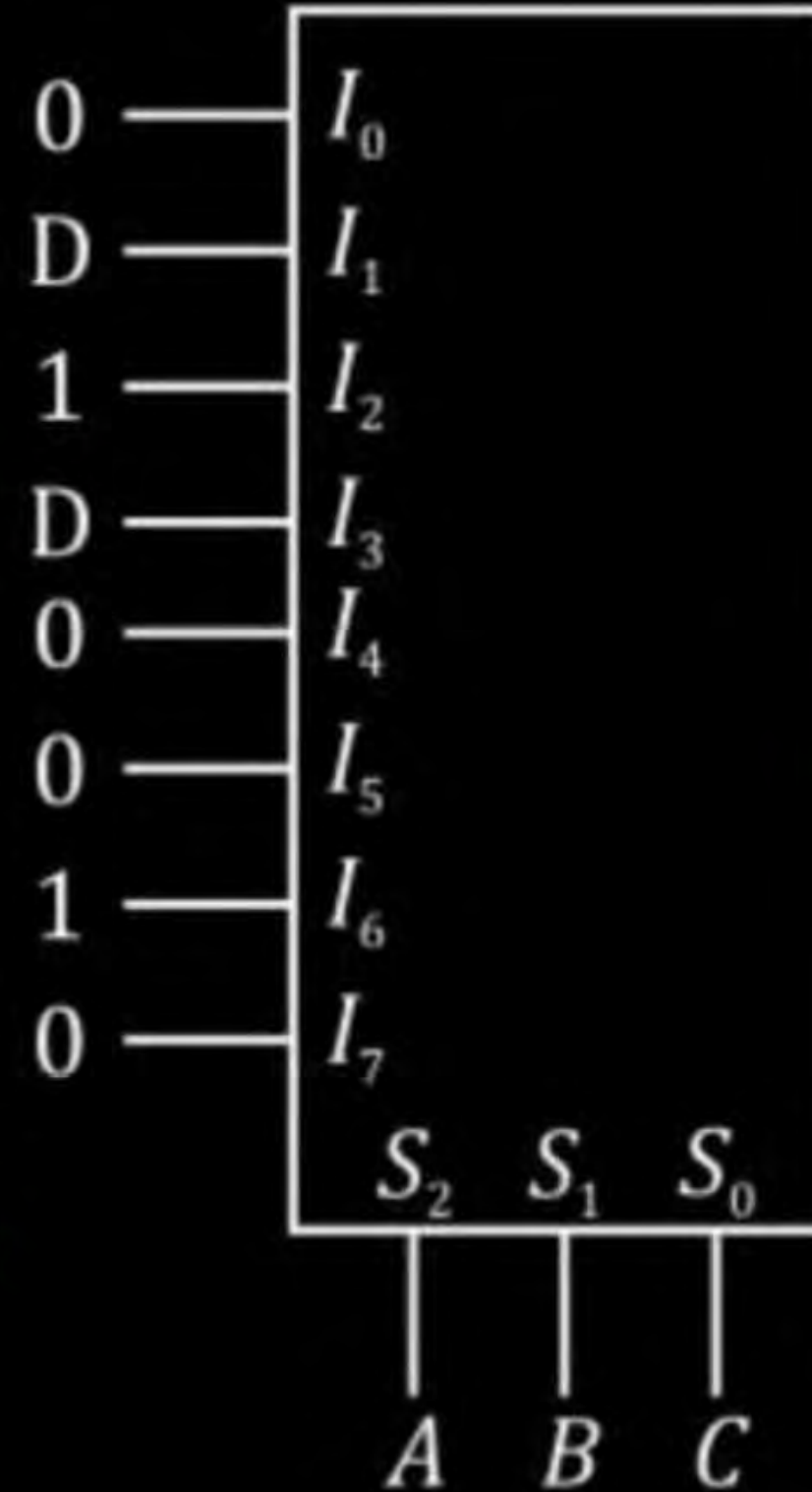
(b) ~~$BC + A\bar{B}D$~~

(c) $B\bar{C} + \bar{A}CD$

(d) ~~$\bar{B}C + A\bar{B}D$~~

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$			1	
$\bar{A}B$	1	1	1	
AB	1	1		
$A\bar{B}$				

$\downarrow B\bar{C}$
 $\downarrow \bar{A}CD$



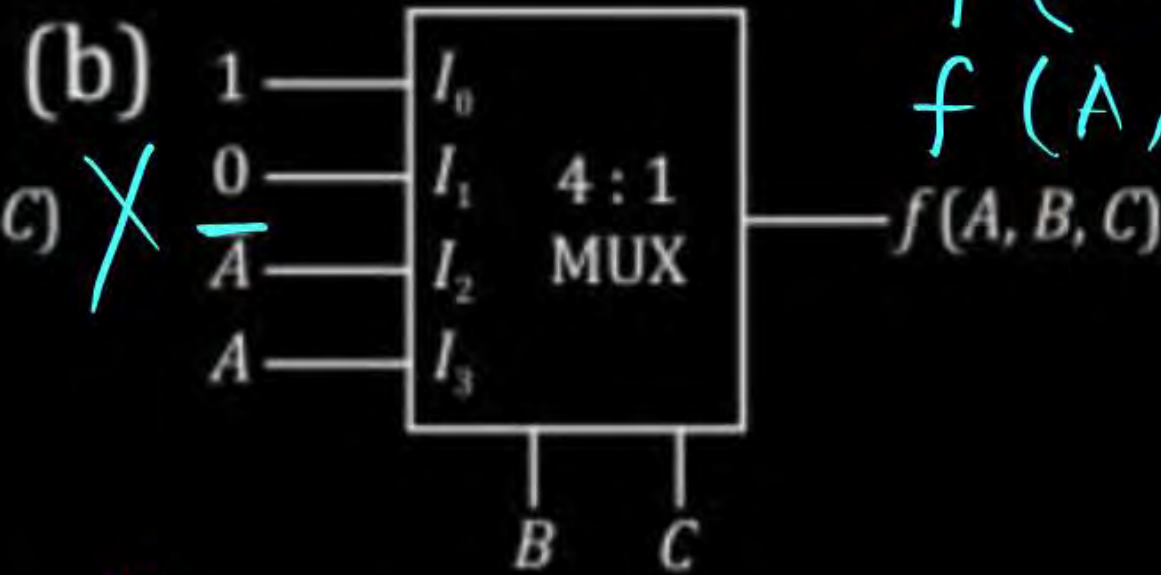
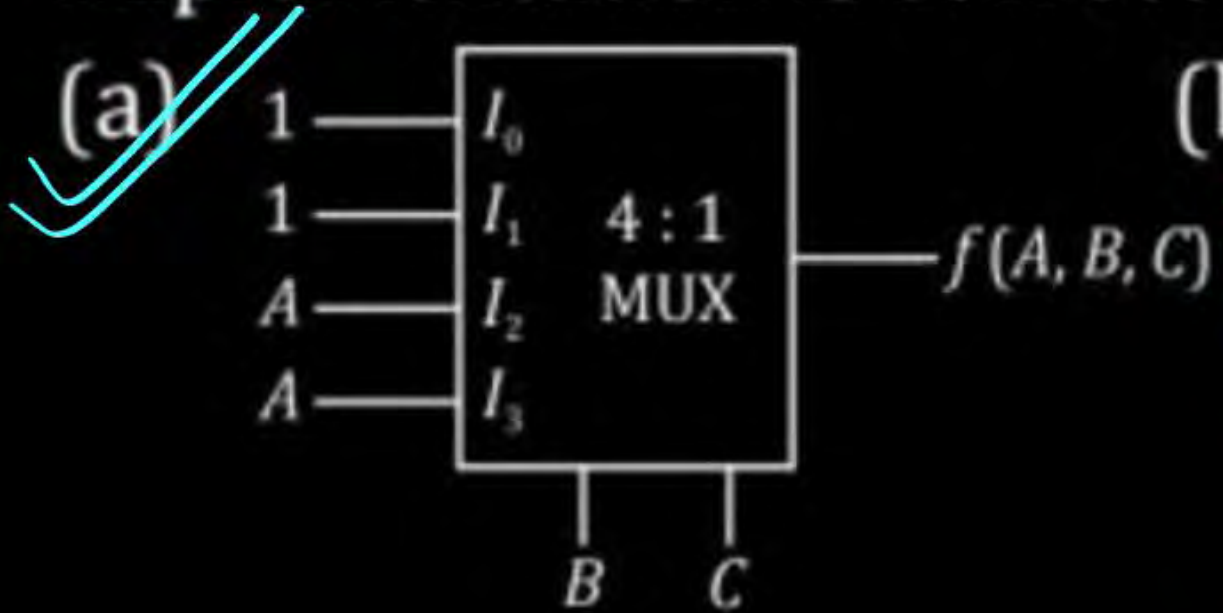
$$\begin{aligned}
 y &= \bar{A}\bar{B}\bar{C} \cdot 0 \\
 &+ \bar{A}\bar{B}C \cdot D \\
 &+ \bar{A}B\bar{C} \cdot 1 + \bar{A}BCD \\
 &+ AB\bar{C} \cdot 1
 \end{aligned}$$

[Question 13]

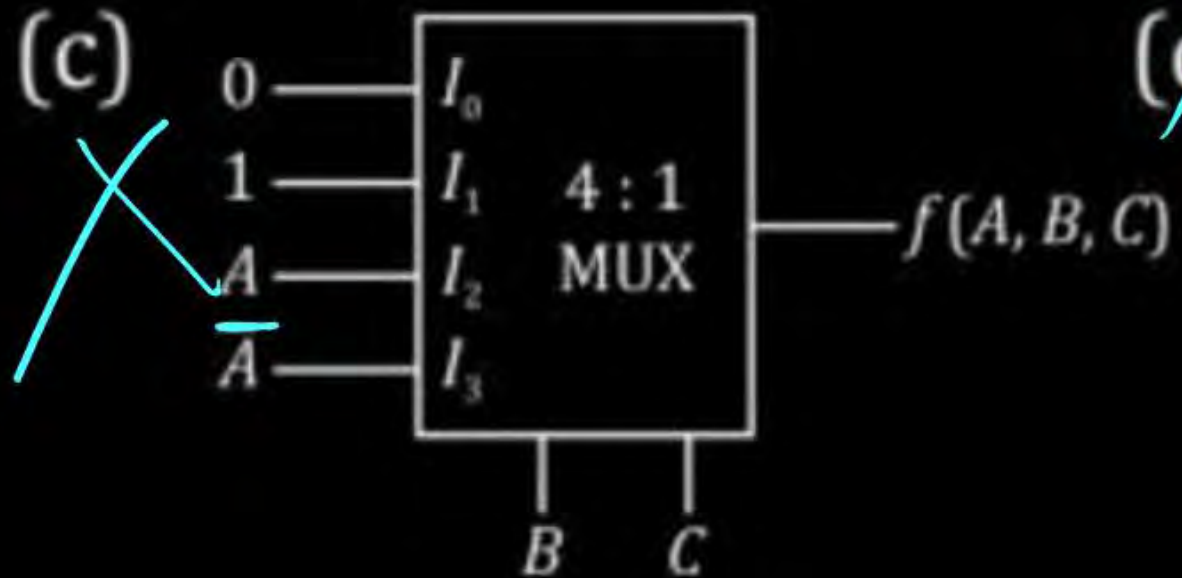


A logical function is given as : $f(A, B, C) = \overline{A}B + B\overline{C} + AB$

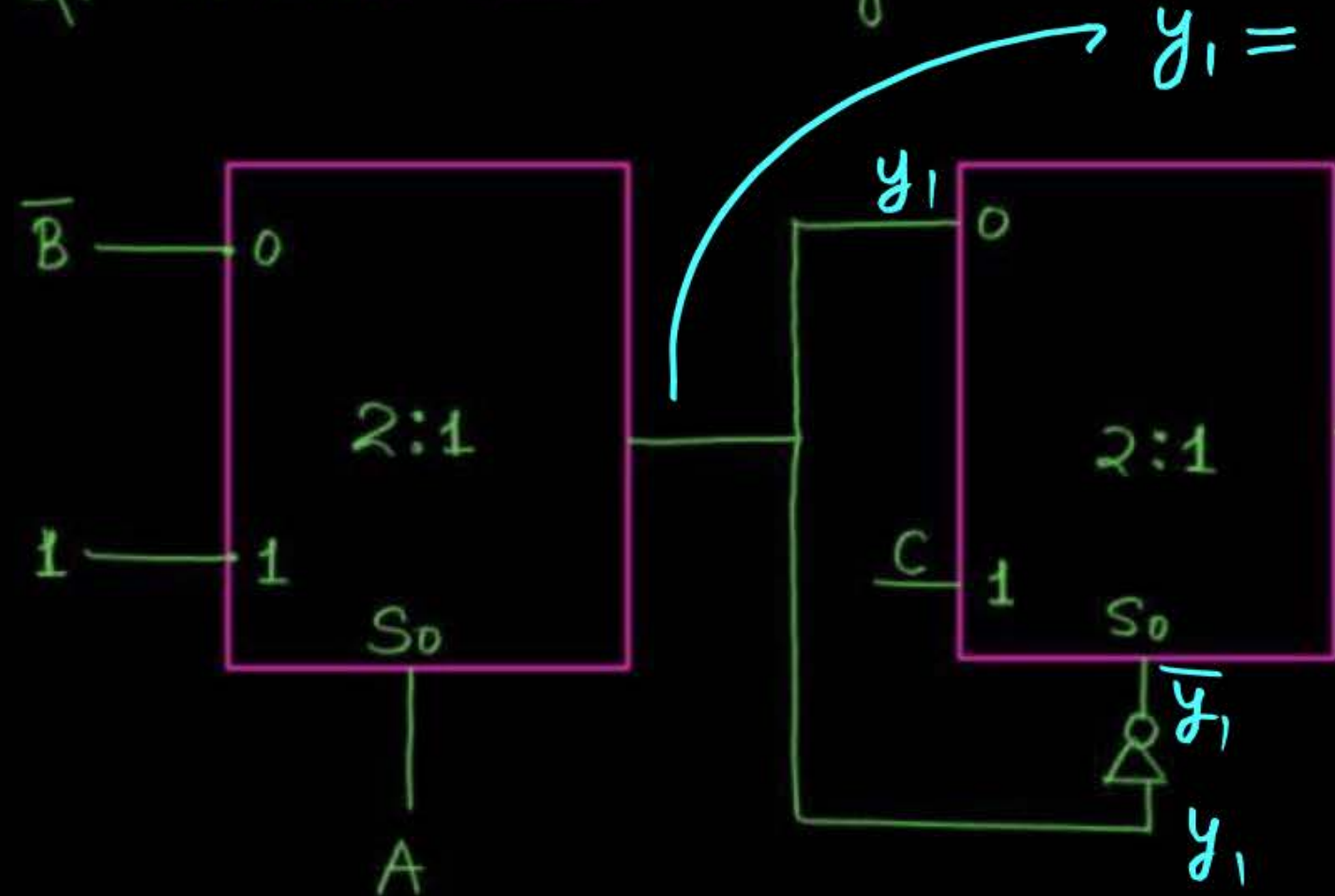
Its implementation using MUX is given below. Then which implementation is correct



$$f(A, 0, 0) = 1, I_0 = 1$$
$$f(A, 0, 1) = 1, I_1 = 1$$



#Q. A MUX CKT is as given below:



$$y_1 = \bar{A} \cdot \bar{B} + A \cdot 1 = (A + \bar{B})$$

$$y = y_1 \cdot y_1 + \bar{y}_1 \cdot C$$

$$= y_1 + \bar{y}_1 \cdot C = y_1 + C$$

$$= (A + \bar{B} + C) = \pi(2)$$

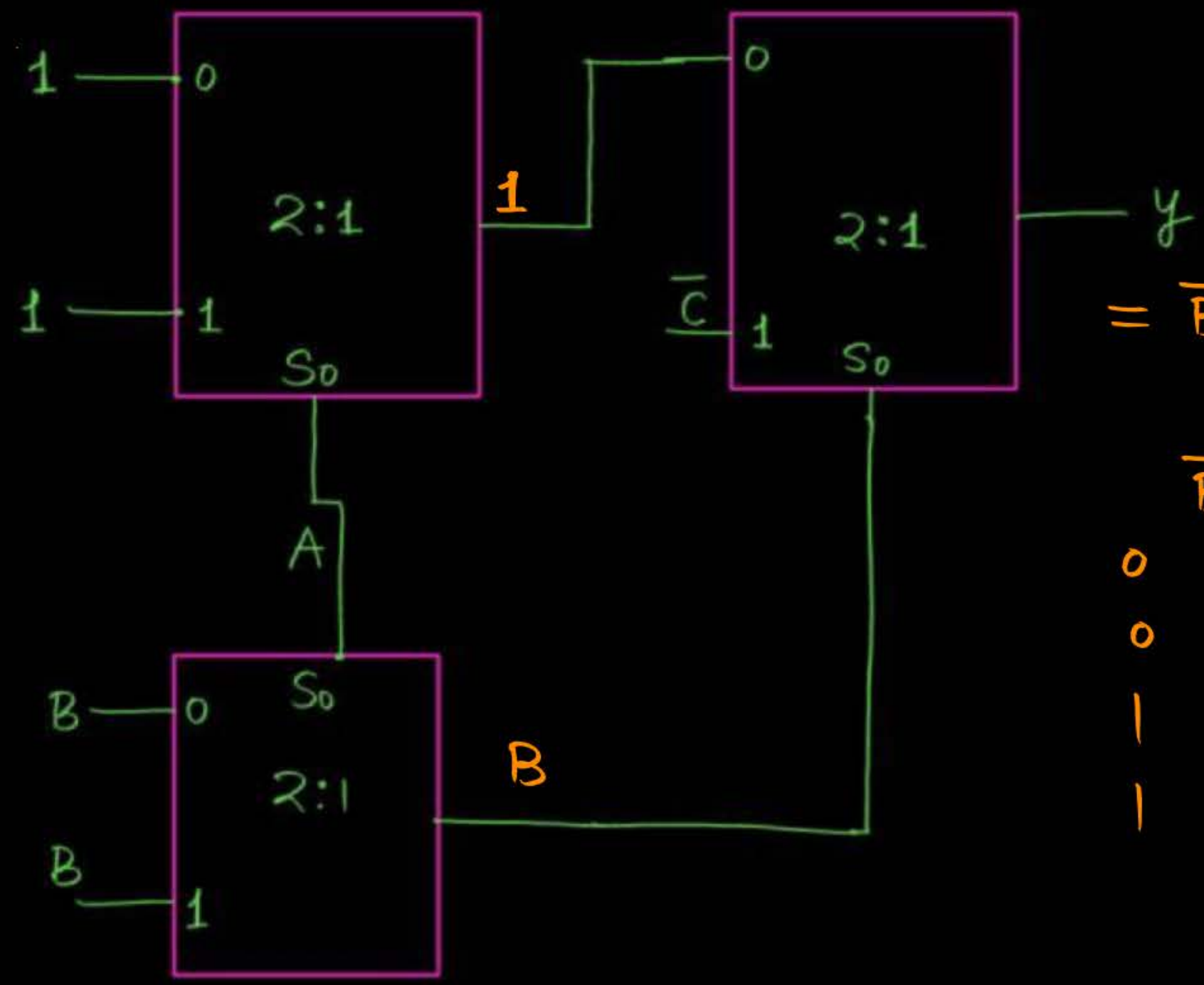
The no. of distinct combination of A, B & C for which o/p y will be 1 is 7.

A	B	C	y
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

$$= \Sigma(0, 1, 3, 4, 5, 6, 7)$$



#Q. A MUX CKT is as given below:



No. of different combination of A, B, C for which O/P y will be 1 6.

$= \Sigma(0, 1, 2, 4, 5, 6)$

$= \overline{B} \cdot 1 + B \cdot \overline{C}$

	$\overline{B}$	
0	0	$0 \rightarrow 0$
0	0	$1 \rightarrow 1$
1	0	$0 \rightarrow 4$
1	0	$1 \rightarrow 5$

	$B \overline{C}$	
0	1	$0 \rightarrow 2$
1	1	$0 \rightarrow 6$

Q. A MUX CKT is as given below:

We want to implement a function
 $f(A, B, C) = \sum(0, 1, 4, 6, 7)$ using
 given MUX CKT.

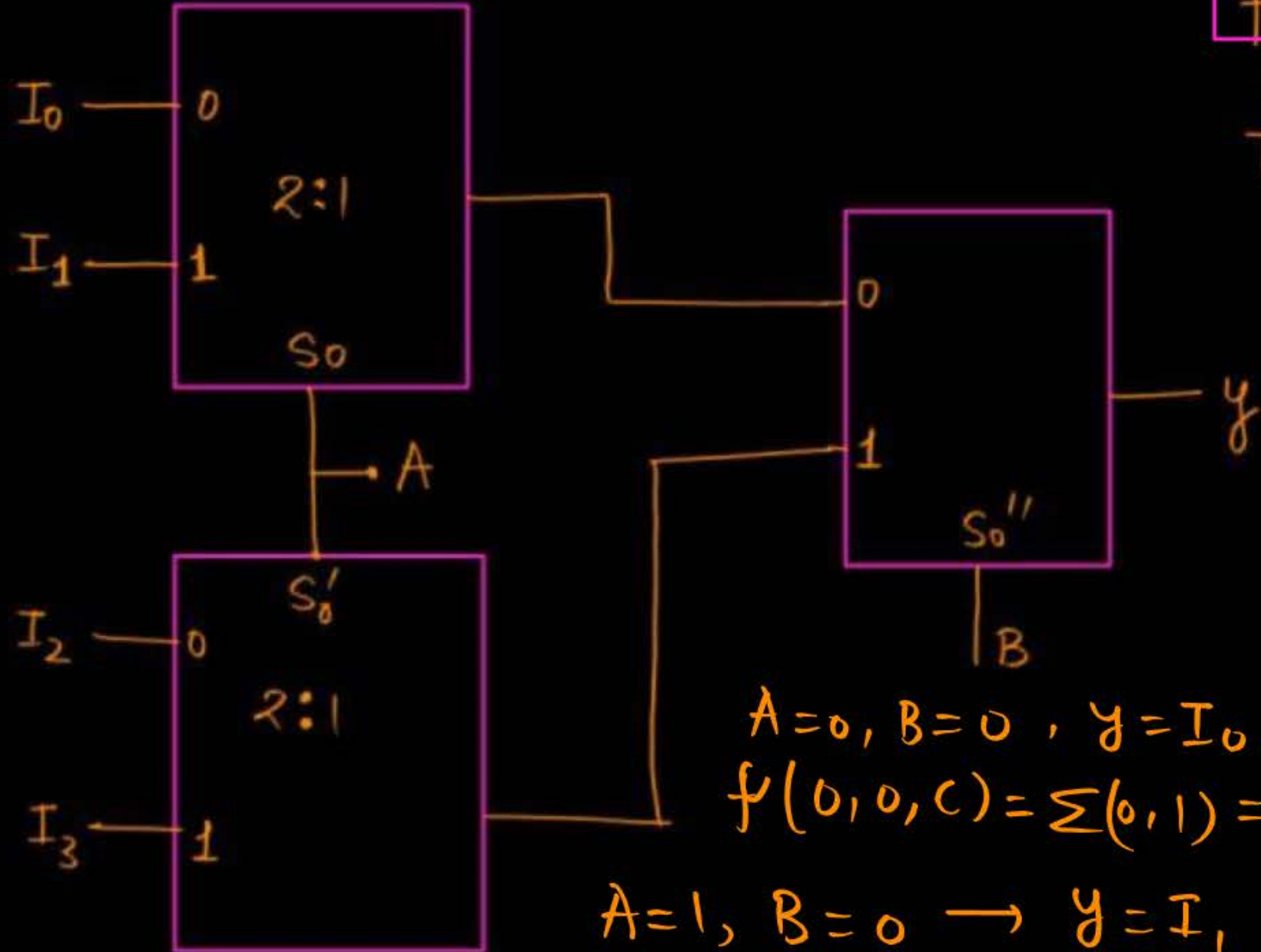
Then i/p lines I_1 & I_2 must be

☒ a. $I_1 = 0, I_2 = \bar{C}$

☒ b. $I_1 = 0, I_2 = C$

☒ c. $I_1 = \bar{C}, I_2 = C$

☒ d. $I_1 = \bar{C}, I_2 = 0$



$A=0, B=0, y=I_0$
 $f(0, 0, C) = \sum(0, 1) = 1$

$A=1, B=0 \rightarrow y=I_1$

$A=0, B=1, y=I_2$

$f(1, 0, C) = \sum(4) = A\bar{B}\bar{C} = \bar{C}, I_1 = \bar{C}$

$f(0, 1, C) = 0, I_2 = 0$

	I_0	I_1	I_2	I_3
$\bar{C}$	(0)	2	(4)	(6)
C	(1)	3	5	(7)
	1	0	$\bar{C}$	1

Wrong

[H.A. and F.A. using 2 : 1 MUX]



- H.A. using 2 : 1 MUX : $\longrightarrow 3$

$$x, y \longrightarrow \boxed{\begin{array}{l} S(x, y) = x \oplus y \\ C(x, y) = xy \end{array}}$$

$$H.S \longrightarrow 3 (2:1)$$

H.W.

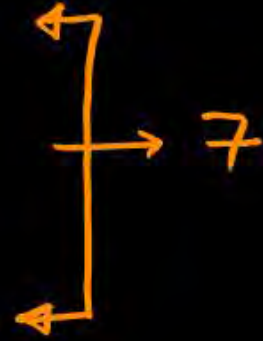


- F.A. using 2 : 1 MUX :

$x, y, z,$

$$S(x, y, z) = x \oplus y \oplus z$$

$$C(x, y, z) = \underbrace{(x \oplus y)} \cdot z + \underbrace{xy} = xy + yz + zx$$



F.S. $\rightarrow 7$

[DeMUX]

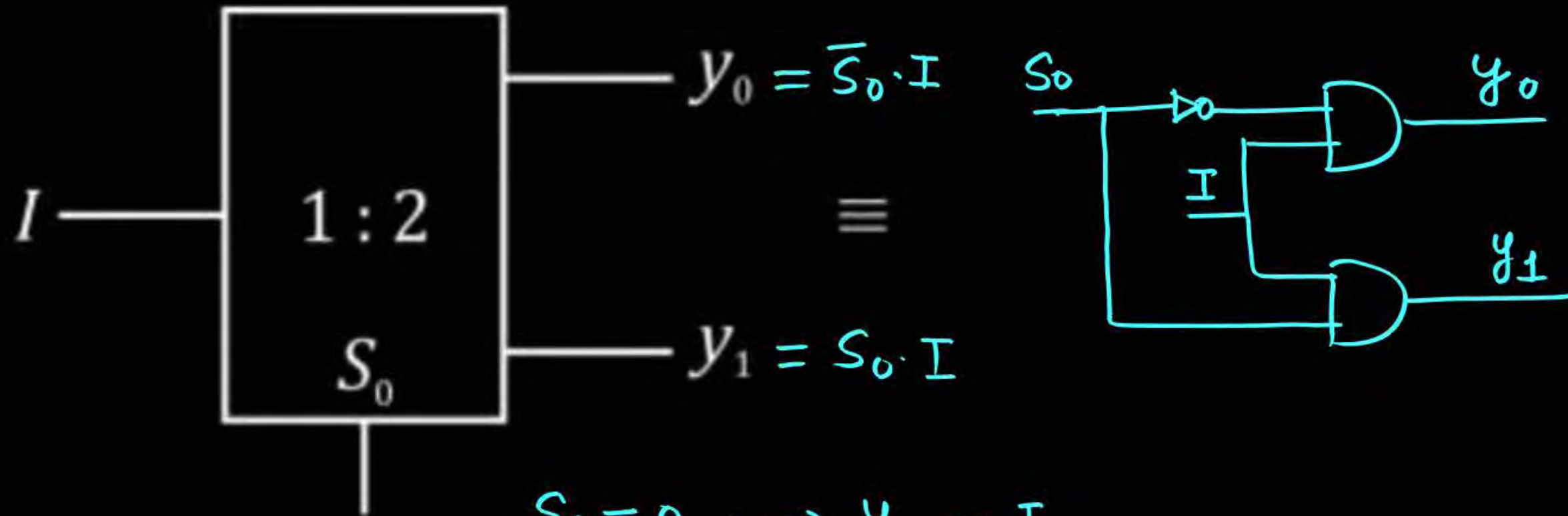


What is DeMUX ?

It is combinational ckt having one i/p line & many o/p line and on the basis of select i/p, i/p line is transferred to one of the o/p line.

- 1 : 2 DeMUX

Internal Circuit



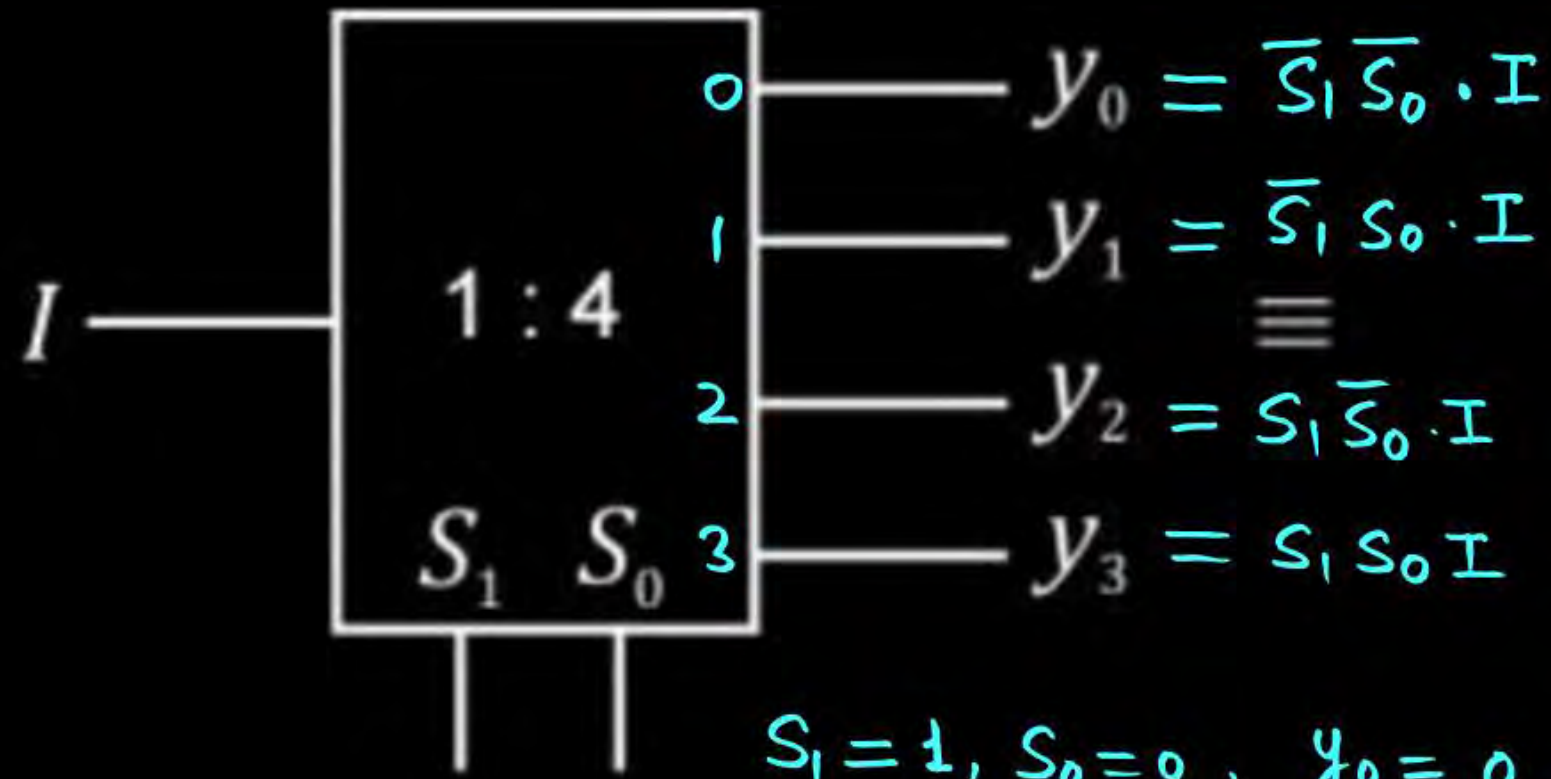
$$S_0 = 0 \rightarrow y_0 = I$$

$$y_1 = 0 \cdot I = 0$$

$$S_0 = 1 \rightarrow y_0 = 0$$

$$y_1 = I$$

- 1 : 4 DeMUX



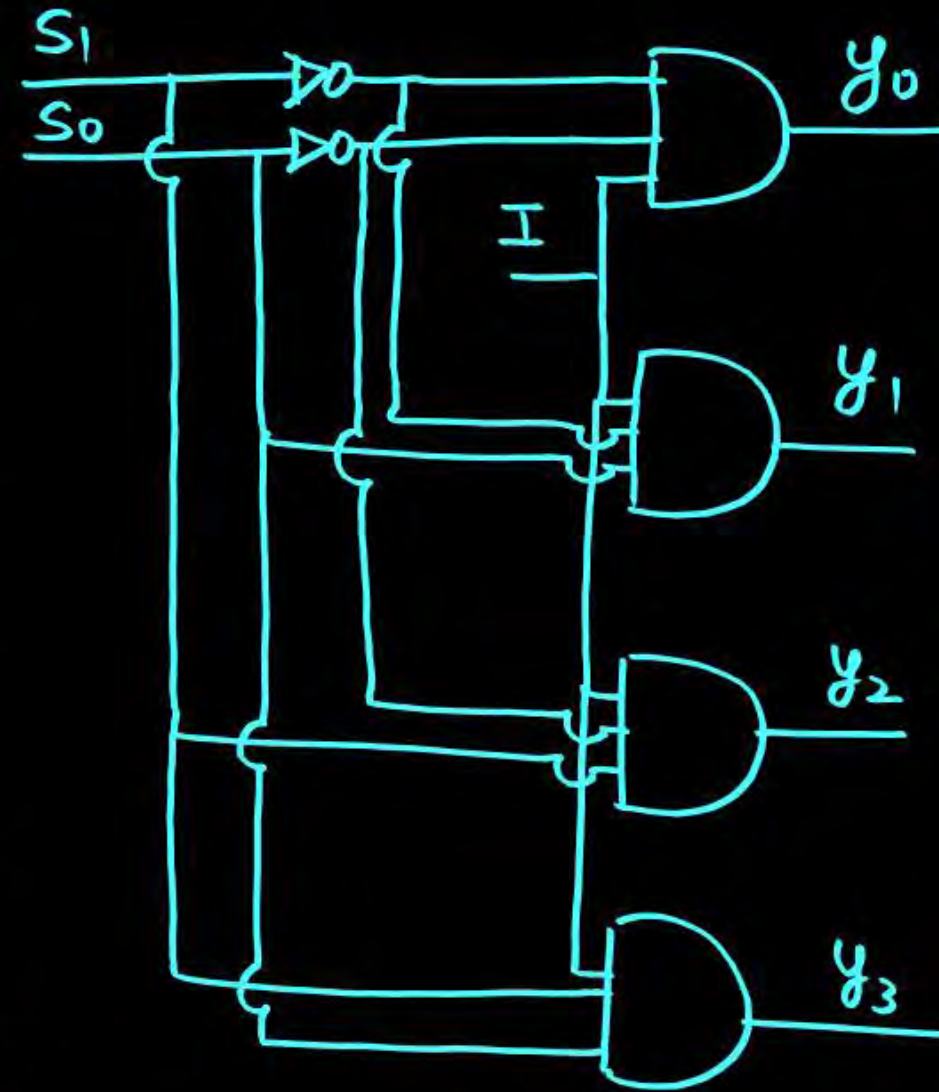
$$S_1 = 1, S_0 = 0, \quad y_0 = 0$$

$$y_1 = 0$$

$$y_2 = I$$

$$y_3 = 0$$

Internal Circuit



[Higher order DeMUX using lower order DeMUX]



- $1:4$ $\xrightarrow{\text{Using } 1:2}$ $2 + 1 = 3 = (2^2 - 1)$
 $1:2^2$
- $1:8$ $\xrightarrow{\text{Using } 1:2}$ $4 + 2 + 1 = 7 = (2^3 - 1)$
 $1:2^3$
- $1:16$ $\xrightarrow{\text{Using } 1:2}$ $8 + 4 + 2 + 1 = 15 = (2^4 - 1)$
 $1:2^4$
- $1:2^n$ $\xrightarrow{\text{Using } 1:2}$ $(2^n - 1)$

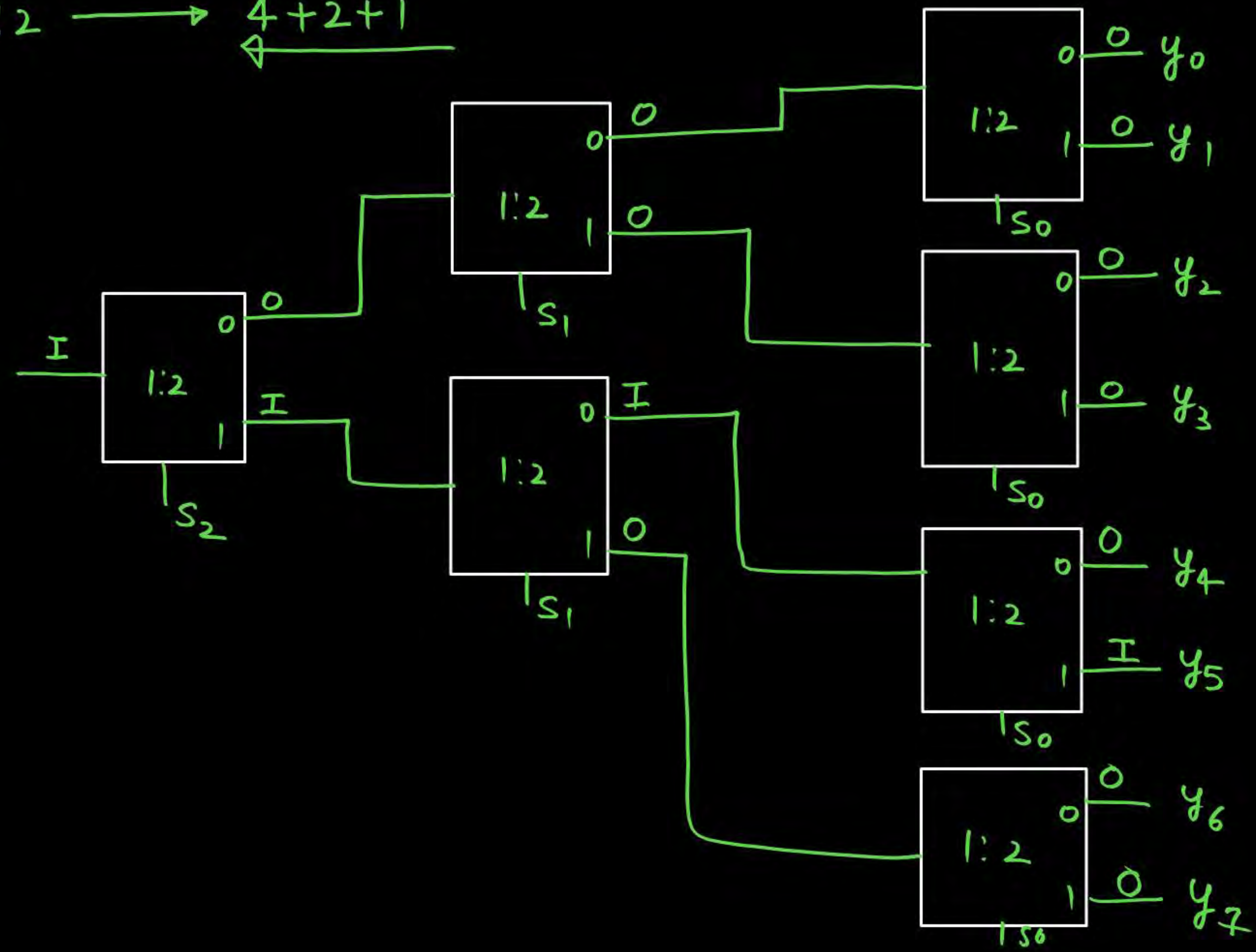
- $1 : 16 \xrightarrow{\text{Using } 1 : 4} 4 + 1 = 5$
- $1 : 32 \xrightarrow{\text{Using } 1 : 4} (8 + 2)$
 $10(1 : 4) + 1(1 : 2)$
- $1 : 64 \xrightarrow{\text{Using } 1 : 8} 8 + 1 = 9$
- $1 : 256 \xrightarrow{\text{Using } 1 : 8} (32 + 4)$
 $36(1 : 8) + 1(1 : 4)$



1:8 using 1:2 $\rightarrow 4+2+1$

$\downarrow$
 I, S_2, S_1, S_0

S_2 S_1 S_0 = 1 0 1



H.W.

1:16 DeMUX using (i) 1:4
(ii) 1:8



2 Minute Summary

- MUX
- DeMUX.

Thank you

GW
Soldiers!

